

Tutorial – 4

1. The employee list for a company contains employee code, name, designation and basic pay. The employee is given a house rent allowance (HRA) of 10% of the basic pay and dearness allowance (DA) of 45% of the basic pay. The total pay of the employee is calculated as Basic Pay + HRA + DA. Write a class to define the details of the employee. Write a constructor to assign the required initial values. Add a method to calculate HRA, DA and total pay and print them. Write another class with main method. Create objects for three different employees and calculate HRA, DA and total pay.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
```

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_4
```

```
{
    class Employee
    {
```

```
        public string EmpCode;
        public string Name;
        public string Designation;
        public double BasicPay;
        public double HRA;
        public double DA;
        public double TotalPay;
```

```
        public Employee(string empCode, string name, string designation, double basicPay)
        {
            EmpCode = empCode;
            Name = name;
            Designation = designation;
            BasicPay = basicPay;
        }
```

```
        // Method to calculate HRA, DA, and total pay
        public void CalPay()
        {
            HRA = 0.10 * BasicPay; // HRA = 10% of basic pay
```

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```
DA = 0.45 * BasicPay; // DA = 45% of basic pay
TotalPay = BasicPay + HRA + DA;
```

```
}
```

```
public void DisplayDetails()
{
```

```
    Console.WriteLine("Employee Code: " + EmpCode);
    Console.WriteLine("Name: " + Name);
    Console.WriteLine("Designation: " + Designation);
    Console.WriteLine("Basic Pay: " + BasicPay);
    Console.WriteLine("HRA: " + HRA);
    Console.WriteLine("DA: " + DA);
    Console.WriteLine("Total Pay: " + TotalPay);
    Console.WriteLine("-----");
}
```

```
}
```

```
internal class Q1
```

```
{
```

```
    static void Main(string[] args)
```

```
    {
```

```
        // Creating objects for three employees
```

```
        Employee emp1 = new Employee("E001", "Mohin", "Manager", 50000);
        Employee emp2 = new Employee("E002", "Rahul", "Developer", 40000);
        Employee emp3 = new Employee("E003", "Saurav", "Analyst", 35000);
```

```
        // Calculate pay and display details
```

```
        emp1.CalPay();
        emp1.DisplayDetails();
```

```
        emp2.CalPay();
        emp2.DisplayDetails();
```

```
        emp3.CalPay();
        emp3.DisplayDetails();
```

```
    }
```

```
}
```

```
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial-4>Q1
Employee Code: E001
Name: Saurav
Designation: Manager
Basic Pay: 50000
HRA: 5000
DA: 22500
Total Pay: 77500
-----
Employee Code: E002
Name: Ravi
Designation: Developer
Basic Pay: 40000
HRA: 4000
DA: 18000
Total Pay: 62000
-----
Employee Code: E003
Name: Anjali
Designation: Analyst
Basic Pay: 35000
HRA: 3500
DA: 15750
Total Pay: 54250
-----
```

2. From the following code and given output, complete missing statements and find out error code and correct it.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
```

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_4
{
    // Base class
    class Shape
    {
        public double Width;
        public double Height;

        public void ShowDim()
```

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```
{
    Console.WriteLine("Width and height are " + Width + " and " + Height);
}

// Derived class
class Triangle : Shape
{
    public string Style; // style of triangle

    // Return area of triangle
    public double Area()    // << Missing statement-1: return type and method signature
    {
        return Width * Height / 2;
    }

    // Display a triangle's style
    public void ShowStyle() // << Missing statement-2: display Style
    {
        Console.WriteLine("Triangle is " + Style);
    }
}

internal class Q2
{
    static void Main()
    {
        // Correct object creation
        Triangle t1 = new Triangle();
        Triangle t2 = new Triangle();

        // Assign values to t1
        t1.Width = 4.0;
        t1.Height = 4.0;
        t1.Style = "isosceles";

        // Assign values to t2
        t2.Width = 8.0;
        t2.Height = 12.0;
        t2.Style = "right";

        // Display info for t1
```

24SOECE13901

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```
Console.WriteLine("Info for t1: ");
t1.ShowStyle();
t1.ShowDim();
Console.WriteLine("Area is " + t1.Area());

Console.WriteLine();

// Display info for t2
Console.WriteLine("Info for t2: ");
t2.ShowStyle();
t2.ShowDim();
Console.WriteLine("Area is " + t2.Area());
    }
}
}
```

Output:

```
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Info for t1:
Triangle is isosceles
Width and height are 4 and 4
Area is 8

Info for t2:
Triangle is right
Width and height are 8 and 12
Area is 48
```

3. Draw a real picture for single level inheritance.

Perform following tasks.

Task 1: Create a class

Task 2: Add few data members as private, protected and public

Task 3: Add few methods as public to work on defined data members

Task 4: Create another applicable class which inherits members from above class

Task 5: Add few data members as private, protected and public into second class

Task 6: Add few methods as public to work on defined data members into second class

Task 7: Create a Demo class with main method.

Task 8: Create at least two objects of a second class defined in Task 4 into main method and call all methods using that object.

Task 9: Write comment for each important portion of code like data members' declaration, methods, some important logic etc.

Task 10: Summarize above solution in your own few words to visualize the solution to the end user.

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```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_4
{
    // Task 1: Create a class
    class Person
    {

        // Task 2: Add data members with different access specifiers
        private int id;
        protected int age;
        public string name;

        // Task 3: Public methods to work with data members
        public void SetDetails(int id, int age, string name)
        {
            this.id = id;
            this.age = age;
            this.name = name;
        }

        public void ShowDetails()
        {
            Console.WriteLine("Person Details:");
            Console.WriteLine("ID:" + id + ", Age:" + age + ", Name: " + name);
        }
    }

    // Task 4: Create another class inheriting Person
    class employee : Person
    {

        // Task 5: Add data members with different access specifiers
        private string empCode;
        protected string department;
        public double salary;
```

```
// Task 6: Public methods to work with data members
public void SetempDetails(string empCode, string department, double salary)
{
    this.empCode = empCode;
    this.department = department;
    this.salary = salary;
}

public void ShowempDetails()
{
    Console.WriteLine("Employee Details:");
    Console.WriteLine("Name" + name + ", Age:" + age + ", EmpCode: " + empCode + ",
Department:" + department + ", Salary: " + salary);
}
}

internal class Q3
{
    static void Main()
    {
        // Task 8: Create two objects of Employee
        employee e1 = new employee();
        employee e2 = new employee();

        e1.SetDetails(101, 25, "Saurav");
        e1.SetempDetails("E001", "IT", 50000);
        e1.ShowDetails();
        e1.ShowempDetails();

        Console.WriteLine();

        e2.SetDetails(102, 30, "Haresh");
        e2.SetempDetails("E002", "HR", 60000);
        e2.ShowDetails();
        e2.ShowempDetails();
    }
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial-4>Q3
Person Details:
ID:101, Age:25, Name: Saurav
Employee Details:
Name:Saurav, Age:25, EmpCode: E001, Department:IT, Salary: 50000

Person Details:
ID:102, Age:30, Name: Haresh
Employee Details:
Name:Haresh, Age:30, EmpCode: E002, Department:HR, Salary: 60000
```

4. From the following code and given output complete missing statements and find out error code and correct it.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
```

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_4
{
    class StaticVar
    {
        public static int num; // static variable

        // Constructor (increments num each time object is created)
        public StaticVar()
        {
            num++;
        }

        // Missing statement: static method to return num
        public static int getNum()
        {
            return num;
        }
    }

    internal class Q4
    {
        static void Main(string[] args)
        {
```


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```
// Creating 3 objects (each constructor call increments num)
new StaticVar();
new StaticVar();
new StaticVar();
```

```
Console.WriteLine("Variable num: {0}", StaticVar.getNum());
```

```
Console.ReadKey();
```

```
}
```

```
}
```

```
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial-4>Q4
Variable num: 3
```

5. Find out error code and correct it. Print appropriate output as desired.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
```

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_4
```

```
{
```

```
    public class A // Base class
```

```
    {
```

```
        public A(int value)
```

```
        {
```

```
            // Executes some code in the constructor
```

```
            Console.WriteLine("Base constructor A()");
```

```
        }
```

```
    }
```

```
    public class B : A // Derived class
```

```
    {
```

```
        public B(int value) : base(value)
```

```
        {
```

```
            // Base constructor is called first
```

```
            // Then this code is executed
```

```
            Console.WriteLine("Derived constructor B()");
```

24SOECE13901

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```
}  
}  
internal class Q5  
{  
    static void Main()  
    {  
        // Create base class object  
        A a = new A(0);  
  
        // Create derived class object  
        B b = new B(1);  
    }  
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial-4>Q5  
Base constructor A()  
Base constructor A()  
Derived constructor B()
```

6. Find out error code and correct it. Print appropriate output as desired.

```
using System;  
using System.Collections.Generic;  
using System.Linq;  
using System.Text;  
using System.Threading.Tasks;  
  
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_4  
{  
    abstract class Test  
    {  
        protected int a;          // protected so derived classes can access it  
  
        public abstract void A();  // must have access modifier  
    }  
  
    class Example1 : Test  
    {  
        public override void A()  
        {  

```

24SOECE13901

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```
        Console.WriteLine("Example1.A");
        a++;
        Console.WriteLine("a = " + a);
    }
}

class Example2 : Test
{
    public override void A()
    {
        Console.WriteLine("Example2.A");
        a--;
        Console.WriteLine("a = " + a);
    }
}

internal class Q6
{
    static void Main()
    {
        Test test1 = new Example1();
        test1.A();

        Test test2 = new Example2();
        test2.A();
    }
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial-4>Q6
Example1.A
a = 1
Example2.A
a = -1
```

7. Refer given output and find out error code and correct it.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
```

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```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_4
{
    sealed class AB
    {
        public int x;
        public int y;
    }
    internal class Q7
    {
        static void Main()
        {
            AB sc = new AB();
            sc.x = 110;
            sc.y = 150;

            Console.WriteLine("x = {0}, y = {1}", sc.x, sc.y);
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial-4>Q7
x = 110, y = 150
```

8. This program will throw an exception. Add try, catch and finally blocks to handle this exception.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_4
{
    internal class Q9
    {
        public static void Main()
        {
            int x = 0;
            int div = 0;
```

```
try
{
    div = 100 / x;
    Console.WriteLine("Result: " + div);
}
catch (DivideByZeroException ex)
{
    Console.WriteLine("Error: Cannot divide by zero.");
    Console.WriteLine("Exception details: " + ex.Message);
}
finally
{
    Console.WriteLine("Program execution completed (finally block executed).");
}
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial-4>Q9
Error: Cannot divide by zero.
Exception details: Attempted to divide by zero.
Program execution completed (finally block executed).
```

9. Arrange the code to get desirable output

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_4
{
    class MyException : Exception
    {
        // Constructor that accepts a string message
        public MyException(string message) : base(message)
        {
        }
    }
}
```

```
internal class Q10
{
    public static void Main()
    {
        try
        {
            throw new MyException("my exception generated.");
        }
        catch (Exception e)
        {
            Console.WriteLine("Exception caught here: " + e.Message);
        }
        Console.WriteLine("LAST STATEMENT");
    }
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial-4>Q10
Exception caught here: my exception generated.
LAST STATEMENT
```